



**Department of Education**

**STEM Education  
National Schools of Excellence  
Physics Syllabus  
Grades 12  
Term 3**

| <b><u>DOMAIN</u></b> | <b><u>Phyla</u></b> |
|----------------------|---------------------|
| <b>Science</b>       | <b>Physics</b>      |
| <b>Technology</b>    |                     |
| <b>Mathematic</b>    |                     |

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## SECRETARY'S MESSAGE

This PHYSICS Syllabus is to be used exclusively by teachers of PHYSICS to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM PHYSICS Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in PHYSICS before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating PHYSICS as a fundamental field of science.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of PHYSICS is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of PHYSICS to human lives, including good health and environment.

In order to ensure Teachers, discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for PHYSICS to be used for the Grade 11 STEM Education Program in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD  
Secretary for Education

# INTRODUCTION

The PHYSICS Syllabus introduced to the Schools of Excellence as a critical component of the STEM Education is mostly concerns

Teaching PHYSICS requires laboratory sessions as an essential part of the teacher guide. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of PHYSICS.

The Grade 11 PHYSICS Syllabus entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since PHYSICS is a highly specialized subject students undertaking PHYSICS are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of PHYSICS through the process of imparting basic PHYSICS knowledge and skills to student

| THEMA                   | MODULE                                | TOPICS                                                                                                    |
|-------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------|
| ARTIFICIAL INTELLIGENCE | 12.1 Data and Artificial Intelligence | 1. Introduction to Artificial Intelligence<br>2. Machine Learning<br>3. Neural Networks<br>4. Data Mining |
| ROBOTICS                | 12.1 Robotics and Electronics         | 1. Introduction to Robotics and Electronics<br>2. Control Systems<br>3. Circuit Boards                    |

# RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honorable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

The PHYSICS was introduced as a Bridging Course to the Schools of Excellence. The syllabus introduced the students to the PHYSICS Domain and expects them to develop an appreciation of the discipline in a broader context. In this teacher guide students are introduced to advanced concepts in PHYSICS as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program and the Grades 11 and 12 Syllabi, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

## **STEM Domain**

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, PHYSICS and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

### **Science**

It is the study of the natural world, including the laws of the nature associated with Physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the PHYSICS design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

### **Technology**

It comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and PHYSICS and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

### **Engineering**

It concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in PHYSICS design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. PHYSICS utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the PHYSICS Domain – Civil, Mechanical and Electrical.

### **Mathematics**

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in PHYSICS. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

## STEM PHYSICS CONTENT OVERVIEW – GRADE 12, TERM 3

**Table 1.1: Sequencing & Benchmarking STEM PHYSICS Content Standards**

| Setting STEM PHYSICS Content Standards for Progressive Learning and Assessment National STEM Assessment |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                          |                                                                               |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| DOMAIN: PHYSICS                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                          |                                                                               |
| Thema: Artificial Intelligence                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Broad Content Standard:                                                                                                                                                                                                                                  |                                                                               |
| Module 12.1: Data and Artificial Intelligence                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>12.1a:</b> Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining<br><b>12.1b:</b> Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence |                                                                               |
| Topic                                                                                                   | Performance Indicators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Module Benchmark                                                                                                                                                                                                                                         | Assessment Tasks                                                              |
| 1. Introduction to Artificial Intelligence                                                              | <b>12.1.1a:</b> Define Artificial Intelligence<br><b>12.1.1b:</b> List branches of AI and explain their role in the development of AI<br><b>12.1.1c:</b> Provide example applications of Artificial Intelligence in Health Care, Robotics and Navigation                                                                                                                                                                                                                                               | Students' ability to understand the knowledge and skills of: <ul style="list-style-type: none"> <li>Machine Learning</li> <li>Neural Networks</li> <li>Data Mining</li> </ul>                                                                            | <b>Group Presentation/Research + Written Quiz + Written Test + Assignment</b> |
| 2. Machine Learning                                                                                     | <b>12.1.2a:</b> Research on the different type of Machine Learning<br><b>12.1.2b:</b> Conduct a Presentation based on their research on the different types of Machine Learning<br><b>12.1.2c:</b> Define Machine Learning<br><b>12.1.2d:</b> Define Deep Learning                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                          |                                                                               |
| 3. Neural Networks                                                                                      | <b>12.1.3a:</b> Relate Neurobiology to Artificial Intelligence Neural Networks<br><b>12.1.3b:</b> Explain the theoretical foundations of Neural Networks<br><b>12.1.3c:</b> Research and present explanation and an example on Supervised Training in Neural Networks<br><b>12.1.3d:</b> Research and present explanation and an example on Unsupervised Training in Neural Networks<br><b>12.1.3e:</b> Research and present explanation and an example on Self-Supervised Training in Neural Networks |                                                                                                                                                                                                                                                          |                                                                               |



|                |                                                                                                                                                                                                                                                            |  |  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| 4. Data Mining | <b>12.1.4a:</b> Define and explain data mining<br><b>12.1.4b:</b> Describe the processes involved in data mining<br><b>12.1.4c:</b> Explain the techniques of data mining<br><b>12.1.4d:</b> Conduct Research on the different applications of data mining |  |  |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|

| Thema: Robotics                             |                                                                                                                                                                                                                                                                                                                                                                    | Broad Content Standard:                                                                                                                                                                                     |                                                                                                                  |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Module 12.1: Robotics and Electronics       |                                                                                                                                                                                                                                                                                                                                                                    | <b>12.1a:</b> Assemble a PCB<br><b>12.1b:</b> Design a PCB<br><b>12.1c:</b> Assemble a robot                                                                                                                |                                                                                                                  |
| Topic                                       | Performance Indicators                                                                                                                                                                                                                                                                                                                                             | Module Benchmark                                                                                                                                                                                            | Assessment Tasks                                                                                                 |
| 1. Introduction to Robotics and Electronics | <b>12.2.1a:</b> Define and explain what robotics is<br><b>12.2.1b:</b> Describe the purpose of robotics<br><b>12.2.1c:</b> Highlight and explain the main components of a robot<br><b>12.2.1d:</b> Describe the main functions of a robot<br><b>12.2.1e:</b> Apply knowledge of robotics to its applications                                                       | Students' ability to understand the knowledge and skills of: <ul style="list-style-type: none"> <li>• Robotic Control Systems</li> <li>• Circuit Boards</li> <li>• Assembling and design a robot</li> </ul> | <p style="text-align: center;"><b>Group Presentation/Research + Written Quiz + Written Test + Assignment</b></p> |
| 2. Control Systems                          | <b>12.2.3a:</b> Explain control systems and give examples<br><b>12.2.3b:</b> Describe how control systems work<br><b>12.2.3c:</b> Design a simple control system                                                                                                                                                                                                   |                                                                                                                                                                                                             |                                                                                                                  |
| 3. Circuit Boards                           | <b>12.2.3a:</b> Define and explain circuit boards<br><b>12.2.3b:</b> Describe the functions of circuit boards<br><b>12.2.3c:</b> Explain the different types of circuit boards<br><b>12.2.3d:</b> Highlight the importance of circuit boards<br><b>12.2.3e:</b> Explain how circuit boards work<br><b>12.2.3f:</b> Apply theory and create an actual circuit board |                                                                                                                                                                                                             |                                                                                                                  |

**Table 1.2: Mapping Content Progression by Modules**

| Thema/Module                                                       | Content Standard                                                                                                                                                                                                                                         | 12.1 | 12.1 |
|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| ARTIFICIAL INTELLIGENCE / 12.1<br>Data and Artificial Intelligence | <b>12.1a:</b> Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining<br><b>12.1b:</b> Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence | ✓    |      |
| ROBOTICS / 12.1<br>Robotics and Electronics                        | <b>12.1a:</b> Assemble a PCB<br><b>12.1b:</b> Design a PCB<br><b>12.1c:</b> Assemble a robot                                                                                                                                                             |      | ✓    |

# **THEMA: ARTIFICIAL INTELLIGENCE (AI)**

## Introduction

Artificial Intelligence (AI) refers to the intelligence demonstrated by machines rather than natural intelligence expressed by humans or animals. This mimicked intelligence by machines refers to the cognitive functions, problem-solving methodologies and behavior associated particularly with humans and this is the common perception that is expressed by most people. In this sense, AI gleaned its inspiration from the human brain.

In 1956, the academic discipline of Artificial Intelligence was founded and established by a group of experts from different disciplines (including Mathematics, Engineering, Psychology and Political Science) that sought to discover the possibilities of creating an artificial brain. As such, AI research had explored and rejected a variety of methodologies, including brain simulation and problem-solving modeling. In the first two decades of the twenty-first century, highly mathematical and statistical modelling dominated the subject and created useful applications such as search engines, recommendation systems, and self-driving cars. As technologies improved with research and resources to develop more capable machines became readily available, actions that were previously called "intelligent" were gradually being excluded from AI definitions.

AI technologies are currently used for a variety of applications including robotics for space and underwater exploration, an automated assembly line of robots in large manufacturing industries, and auto correct and face recognition on smartphones. As AI research and technology improves, it is becoming more accessible and integral to the common every-day human being and could become a predominant tool in resolving almost all of humanities problems. This would also suggest that there will be a lesser dependency on humans themselves or human intervention to solve our own problems. This has planted doubts into AI skeptics, developers, and companies on the uncertainties of AI technologies with concerns of it being a possible threat to human survival which is raising many controversial debates today. However so, it is an undisputable fact that AI has improved and aided in the design of the modern world as we know it today.

It is important that students are to be introduced to the concept of AI in preparation of its vast applications in the future. With this introduction, students should be thinking about how they can become creators and innovators in the field of AI and AI development, and not only end-users.

## MODULE 12.1: DATA AND ARTIFICIAL INTELLIGENCE

**4 weeks**

The field of Artificial Intelligence is deeply rooted in using data and information. Based on how its model is created, the data is manipulated to develop “reasoning” that allows the AI to perform its intended function. This is the general field of AI that then diverges into a variety of sub-fields that are categorized according to certain distinct objectives and methods. In this *Thema: Artificial Intelligence*, we will explore and discuss some of these sub-fields and focus on some of the current practices used in AI.

### Linking Modules

- Technology / Grade 12 / Thema Artificial Intelligence / Module 12.1
- Engineering / Grade 12 / Thema Artificial Intelligence / Module 12.1
- Chemistry / Grade 12 / Thema Artificial Intelligence / Module 12.1

### Broad Content Standard

**12.1a:** Conceptualize the working principle of AI via the integration and application of Machine Learning, Neural Networks and Data Mining

**12.1b:** Ultimately generate Innovative ideas regarding Applications of Artificial Intelligence

### Content Benchmarks

Students’ ability to understand the knowledge and skills of:

- Machine Learning
- Neural Networks
- Data Mining

## Topic 1: Introduction to Artificial Intelligence

### Performance Indicators

Students are expected to:

- 12.1.1a:** Define Artificial Intelligence
- 12.1.1b:** List branches of AI and explain their role in the development of AI
- 12.1.1c:** Provide example applications of Artificial Intelligence in Health Care, Robotics and Navigation

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Introduction
  - Defining AI
- Branches of Artificial Intelligence
  - Robotics
  - Machine Learning
  - Deep Learning
  - Expert Systems
  - Fuzzy Logic
  - Natural Language Processing (NLP)
- Applications of Artificial Intelligence
  - Navigation – Autonomous Vehicles
  - Health Care
  - Robotics

#### ***Set of Skills for Proficiency Levels***

- Research Skills – able to research on complex topics
- Relational Skills – ability to relate different concepts

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate AI and its applications

## Topic 2: Machine Learning

### Performance Indicators

Students are expected to:

- 12.1.2a:** Research on the different type of Machine Learning
- 12.1.2b:** Conduct a Presentation based on their research on the different types of Machine Learning
- 12.1.2c:** Define Machine Learning
- 12.1.2d:** Define Deep Learning

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Relation between AI and Machine Learning
  - Machine Learning
  - Deep Learning
  - What is Machine Learning?
  - Uses of Machine Learning
  - Types of Machine Learning
    - Supervised Learning
    - Unsupervised Learning
    - Reinforcement Learning

#### ***Set of Skills for Proficiency Levels***

- Research Skills – able to research on complex topics
- Relational Skills – ability to relate different concepts

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate neural networks as a branch of AI and its applications

## Topic 3: Neural Networks

### Performance Indicators

Students are expected to:

- 12.1.3a:** Relate Neurobiology to Artificial Intelligence Neural Networks
- 12.1.3b:** Explain the theoretical foundations of Neural Networks
- 12.1.3c:** Research and present explanation and an example on Supervised Training in Neural Networks
- 12.1.3d:** Research and present explanation and an example on Unsupervised Training in Neural Networks
- 12.1.3e:** Research and present explanation and an example on Self-Supervised Training in Neural Networks

### Content Expansion

#### *Body of Knowledge for Understanding*

- Introduction
  - Neurobiology
  - Mathematical Theory
  - Theoretical Foundations of Neural Networks
  - A Biological Inspiration
  - Artificial Networks
  - How a Neural Network Learns
    - Supervised Training
    - Unsupervised Training
    - Self-supervised Training
  - Layering of Neural Networks
- Neural Network and Other Information Processing Approaches
  - Overview of Brain Physiology
  - Recent Results in Neurobiology
    - Neurons
    - Synapses
    - Map Representation
    - Self-Organization
    - Network Circuitry
    - Network Dynamics
    - Learning
    - Attention
    - Organizing Principles
- Tasks Neural Networks Perform and Representative Models
  - Introduction
  - Tasks that Neural Networks Perform
    - Classification
    - Self-Organization/Category Formation
    - Associative Memory
    - Sensory Data Processing



- Computational Problems
- Nonlinear Mapping
- Multi-sensor Automata
- Illustrative Neural Network Models
- Available Implementation Tools
- A Simulated World
- Typical Application – The Kanji Character Recognizer
- Typical Application – Multilayer Perception Vowel Classifier

***Set of Skills for Proficiency Levels***

- Research Skills – able to research on complex topics
- Relational Skills – ability to relate different concepts

***Traits of Attitudes and Values to govern their lives***

- Appreciate machine learning as a branch of AI and its applications

## Topic 4: Data Mining

### Performance Indicators

Students are expected to:

- 12.1.4a:** Define and explain data mining
- 12.1.4b:** Describe the processes involved in data mining
- 12.1.4c:** Explain the techniques of data mining
- 12.1.4d:** Conduct Research on the different applications of data mining

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Introduction
- What Motivated Data Mining and what is its Important
- What is Data Mining
- Data Mining on what kind of Data
  - Relational Database
  - Data Warehouses
  - Transactional Database
  - Advanced Data and Information Systems and Advanced Applications
  - Advanced Database Systems
    - Object-Relational Databases
    - Temporal Database, Sequence Databases, and Time-Series Databases
    - Spatial Databases and Spatiotemporal Databases
    - Text Databases and Multimedia Databases
    - Heterogenous Databases and Legacy Databases
    - Data Streams
    - The World Wide Web
- Data Mining Functionalities – What Kind of Patterns can be mined?
  - Concept/Class Description – Characterization and Discrimination
  - Mining Frequent Patterns, Associations, and Correlations
  - Classification and Predication
  - Cluster Analysis
  - Outlier Analysis
  - Evolution Analysis
- Are all the Patterns Interest?
- Classification of Data Mining Systems

#### ***Set of Skills for Proficiency Levels***

- Apply data mining to analyse enormous volumes of data for new and potentially helpful information.

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate data mining and apply it in discovering new and potentially helpful information from analysing large amounts of data.

### **Application of Knowledge, Skills and Values (AKSA)**

- Computer Science
- Engineers
- Mathematicians
- Chemistry
- Biology
- Statisticians

# THEMA: ROBOTICS

## Introduction

Robotics is a specific yet versatile field of study and is a branch of AI. It utilizes robots (hence the name robotics) that are endowed with the equivalence of human senses including vision, touch, and the ability to sense surrounding conditions such as temperature and pressure to perform simple to complicated tasks.

Robots possess the capabilities to exceed human limitations such as to work longer hours, be accurate and precise in calculations and actions, and to work in hazardous conditions. Much like AI, robotics has a widespread of applications including manufacturing industries, medical healthcare, and military purposes for security and defense intelligence.

There are robots that are designed to accomplish pre-programmed instructions and there are some programmed to make simple decisions. Current robotics research is focused on improving this decision-making process for robots to become self-sufficient; allowing them to make decisions in unconstructed environments.

AI and machine learning algorithms have grown more efficient because of advancements in computing, data transfer, storage, and computer power, making it easier to encode programs for robots, design innovative uses for applications, and minimize energy consumption. Additionally, the improvements in sensor technology and material science improve the hardware of robotics to interface with AI capabilities, ultimately improving the field of robotics.

Although much focus is directed towards product technology (e.g., smartphones, commercial jets, automobiles, solar panels, etc.), process technology (which involves the usage of machines to improve how a good or service is produced, enabling automation) will influence significant changes that will later redefine the scope of modern technology.

## **MODULE 12.1: ELECTRONICS**

*4 weeks*

Robotics is a multidisciplinary field combining computer science and engineering. Robotics is the study of the design, manufacture, operation, and use of robots. Robotics aims to create devices that can support and help humans. Mechanical engineering, electrical engineering, information engineering, mechatronics, electronics, bioengineering, computer engineering, control engineering, software engineering, and mathematics are just a few of the subjects that robotics encompasses.

Robots can be employed in a variety of scenarios and for a variety of objectives, but many are now used in hazardous areas (such as radioactive material inspection, bomb detection and deactivation), manufacturing operations, or in situations where humans are unable to live (e.g., in space, underwater, in high heat, and clean up and containment of hazardous materials and radiation). Robots can take on any shape, but some are designed to seem like people.

This module introduces students to Robotics; it highlights the different aspects of the module electronics and describes the different applications of robotics in the real world.

### **Linking Modules**

NIL

### **Broad Content Standard**

**12.1a:** Assemble a PCB

**12.1b:** Design a PCB

**12.1c:** Assemble a robot

### **Content Benchmarks**

Students' ability to understand the knowledge and skills of:

- Robotic Control Systems
- Circuit Boards
- Assembling and design a robot

## Topic 1: Robotics and Electronics

### Performance Indicators

Students are expected to:

**12.2.1a:** Define and explain what robotics is

**12.2.1b:** Describe the purpose of robotics

**12.2.1c:** Highlight and explain the main components of a robot

**12.2.1d:** Describe the main functions of a robot

**12.2.1e:** Apply knowledge of robotics to its applications

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Introduction
  - Robot
  - Robotics
  - Robotics in AI
- Basic Components of a Robot
  - Body
  - Control System
  - Actuator
  - Sensor
  - Drivetrain
- Uses and Examples
  - Industrial Robots
  - Robots in Research
    - Pathfinder and Sojourner
    - Space Shuttle Robotic Arm – Canadarm
    - Special Purpose Dexterous Manipulator – Dextre
  - Robotics in Education
  - Importance of Robotics

#### ***Set of Skills for Proficiency Levels***

- Apply theory of robotics into future practical activities in building and designing a robot.

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate the use of robotics and use its concepts to design robotic solutions to tackle problems faced in the real world.

## Topic 2: Control Systems

### Performance Indicators

Students are expected to:

**12.2.3a:** Explain control systems and give examples

**12.2.3b:** Describe how control systems work

**12.2.3c:** Design a simple control system

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Control System in Robotics
  - Open Loop System
  - Closed Loop Systems
- Closed Loop Control System
- Control of Manipulators – Linear Control of Manipulators
- Types of Robotics Control Systems
  - Point-to-point System
  - Continuous Path Control System
- Controlling Robots
- Robots vs. Robotics
- Robots Replacing Human Development
- Conclusion

#### ***Set of Skills for Proficiency Levels***

- Apply theory of control systems to designing an actual control system.

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate control systems and design it to function in its respective applications



## Topic 3: Circuit Boards

### Performance Indicators

Students are expected to:

- 12.2.3a:** Define and explain circuit boards
- 12.2.3b:** Describe the functions of circuit boards
- 12.2.3c:** Explain the different types of circuit boards
- 12.2.3d:** Highlight the importance of circuit boards
- 12.2.3e:** Explain how circuit boards work
- 12.2.3f:** Apply theory and create an actual circuit board

### Content Expansion

#### ***Body of Knowledge for Understanding***

- Fundamentals
- Schematic Diagram

#### ***Set of Skills for Proficiency Levels***

- Apply the theory of circuit boards to create an actual circuit board

#### ***Traits of Attitudes and Values to govern their lives***

- Appreciate circuit boards and design it to be used in robotics and other key areas.

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